

Fantastic 4 + 1

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Figure 0.1: Fantastic 4 + 1 Robot, Leonard

ES 51 Turf Wars Competition Team 4
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Abstract

The goal of this project was to design a robot capable of navigating a turf field, collecting designated scoring objects, and strategically placing them onto tiered shelves to maximize points. Important design constraints included fitting our robot within an 11" x 11" x 11" cube, reaching heights up to 21", and incorporating all required fabrication methods including milling, laser cutting, turning, 3D printing, drilling, and molding. The scoring shelves were positioned at 7", 14", and 21" awarding one, two, and three points respectively for placing a wooden block, a tennis ball, and a dog toy on their correct shelves. Our strategy prioritized both time and point efficiency by utilizing our robot's effective drivetrain to quickly navigate to the objects scattered around the field and lifting them onto the correct shelves. To achieve this, we developed a telescoping linear slide mechanism driven by belts wrapped around staged pulleys and paired with a horizontal claw attachment for object placement. The drivetrain, powered by two motors and equipped with four Eco-flex 30 wheels, successfully climbed both 15° and 30° ramps during the drivetrain demo. In the final Turf Wars competition, our robot performed reliably. Although outpaced by the reliable 4-bar linkage designs other teams went with, our robot earned third place overall and was the most successful non-linkage based design.



Figure 0.2: Turf wars field set up

1. Concept Development

1.1 Turf War Competition Overview

The Fall 2025 Turf Wars competition required teams to design robots capable of collecting scoring objects, wooden blocks, tennis balls, and dog toys, and placing them on platforms at heights of 7", 14", and 21". Teams competed in four minute rounds, and the robot with the highest point total at the end advanced.

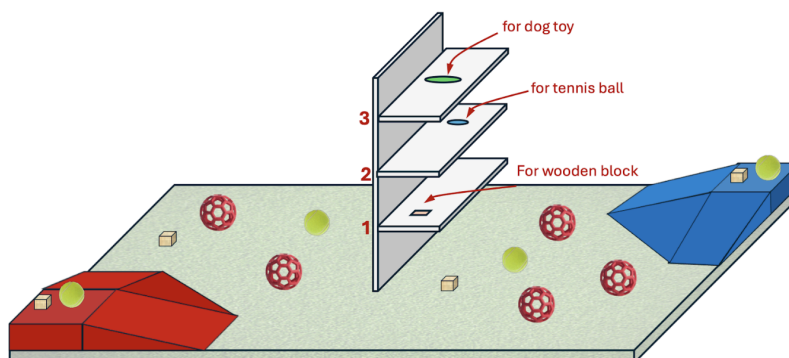


Figure 1.1: Turf wars field set up

1.1a Field Design

The field was covered in turf and was bordered by four ramps located on the corners of the field, with two ramps at a 15° incline two of the ramps at a 30° incline. All robots were required to demonstrate the ability to climb the 15° ramp prior to the start of the competition. At the center of the field stood the three scoring platforms, with the highest platform reaching up to 21".

1.1b Scoring

Scoring was determined by both the object type and placement position of the object on the shelf. A wooden block placed on the 7" shelf earned one point. A tennis ball placed on the 14" shelf earned two points. A dog toy placed on the 21" shelf earned three points. An additional one point was awarded if the object was placed inside of one of the outlined shapes marked on the shelves, meaning that a tennis ball placed on the correct shelf and placed inside the shape outline would earn three points instead of just two.

1.2 Constraints

All robots were required to meet several competition constraints. These constraints shaped the practicality and feasibility of each robot design that was considered.

1.2a Materials

Robots could only be built using materials supplied by the ES51 lab, including ¼" and ⅛" acrylic sheets, aluminum and steel shafts, PLA filament, and standard hardware including fasteners and rotational components located in the inventory space. Total 3D printing volume was restricted to

15 in³. In the case of a tie during competition seeding, the robot with the lower total material cost advanced, incentivising efficient material usage.

1.2b Motors

Each team was limited to a maximum of six actuators, three motors and three servos with a maximum current draw of 9 amps. Motors were characterized in the lab to determine the torque output for design calculations.

1.2c Size

All robots were required to fit within the Box of Justice, a 11" x 11" x 11" cube. After the round began, the robot was allowed to extend larger than this volume

1.2d Time

The final robot had to be fully fabricated, assembled, and tested by impound day, Thursday, December 4th, at 3:00 PM. This hard deadline influenced feasible design complexity and manufacturability.

1.3 Final Design

After evaluating several concepts, the robot our team selected consisted of a telescoping linear slide and horizontal claw attached to a four wheel drivetrain. This design satisfied all material, size and motor constraints while providing the necessary reach and power to lift and place scoring objects on the shelves.

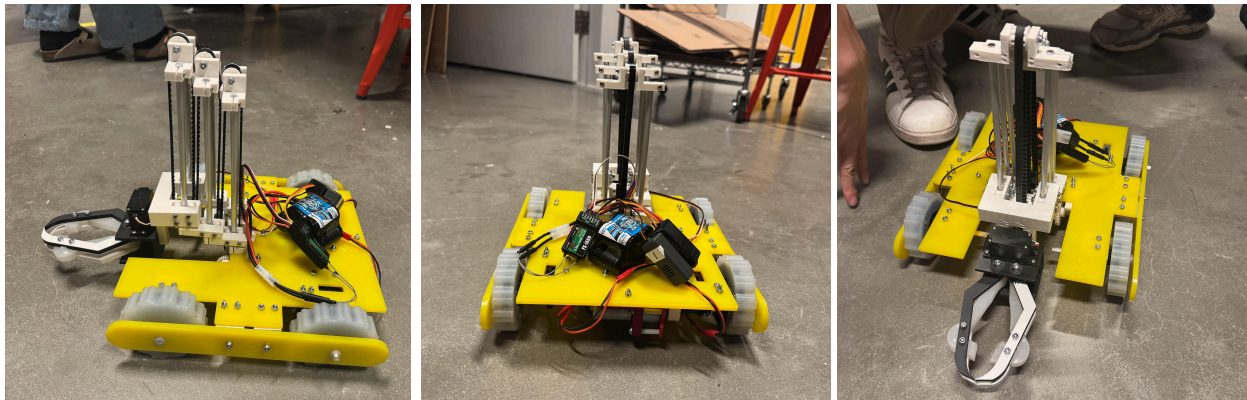


Figure 1.2: Final robot design, viewed from different angles

1.3a Linear Slide

The lifting mechanism of our robot consisted of a three stage telescoping linear slide mechanism. Capable of extending 15" in height. Each stage consisted of 3D printed base and top platforms connected by aluminum shafts, and powered by connected timing belts and pulleys. The mechanism was selected after the evaluation because of its high vertical reach and a compact

retracted length, low lateral movement from its starting position to improve scoring precision, and fast extension relative to other mechanisms that were considered.

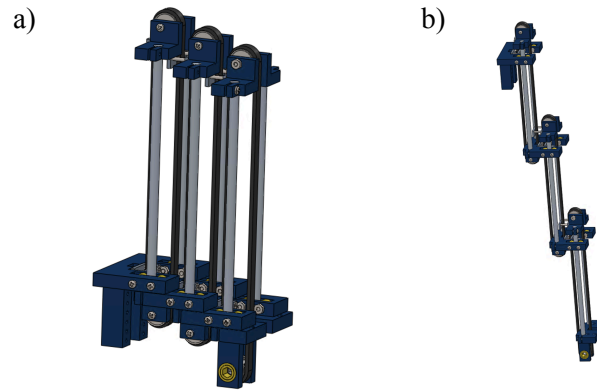


Figure 1.3: Linear Slide CAD, **a)** Retracted State, **b)** Extended State

1.3b Horizontal Claw

Object pick-up was achieved through a servo actuated horizontal claw mechanism mounted to the top stage of the linear slide. Spur gears synchronized the rotation of the two claw prongs, ensuring symmetric closure. To improve grip, a layer of silicone was stretched between two layers of acrylic plates, increasing friction and adaptability for grasping the scoring objects in irregular positions.

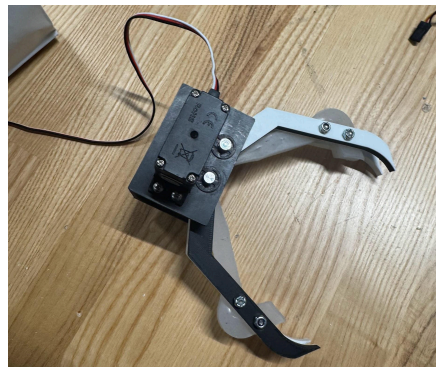


Figure 1.4: Completed Horizontal Claw

1.3c Drivetrain

The drivetrain consisted of a four-wheel U-shaped chassis. The baseplate was initially lasercut with premade holes but was later modified through careful drilling, to add extra holes to accommodate for mounting changes during assembly. The open front space left by the U-shape of the chassis allowed our linear slide be secured lower to the ground and provided the clearance needed to fit within the 11" height constraint.

Rear-wheel drive was selected based on torque calculations and time constraints. The two rear motors supplied sufficient torque to climb both the 15° and 30° ramps during drivetrain demo.

Wheels were cast using Eco-Flex 30 silicone to provide traction and 3" diameter wheels were selected to provide adequate ground clearance while still maintaining a low center of mass.

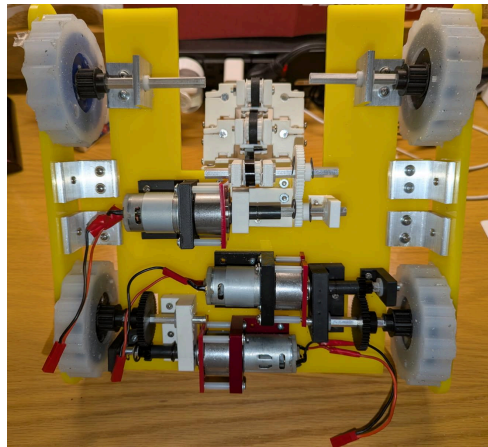


Figure 1.5: Completed Drivetrain

1.4 Alternate Solutions

Prior to selecting our final concept we developed and evaluated two alternative designs. Although neither was ultimately selected, each design helped inform our final decision.

1.4a Scissor Lift Design

This concept featured a front shovel for object collection, a vertical claw, and a multi-stage scissor lift for vertical extension. The design provided the advantages of precise vertical motion and sufficient lift height to score all objects. However, quantitative evaluation using a Pugh matrix revealed several critical drawbacks. The scissor lift required long linkages and multiple pivot joints, increasing the likelihood of mechanical instability. Its extension speed was also limited due to friction at each pivot, and the overall weight distribution was unfavorable, raising the risk of tipping while climbing ramps. Together, these factors made the design unlikely to satisfy the competition's requirements for speed and stability.

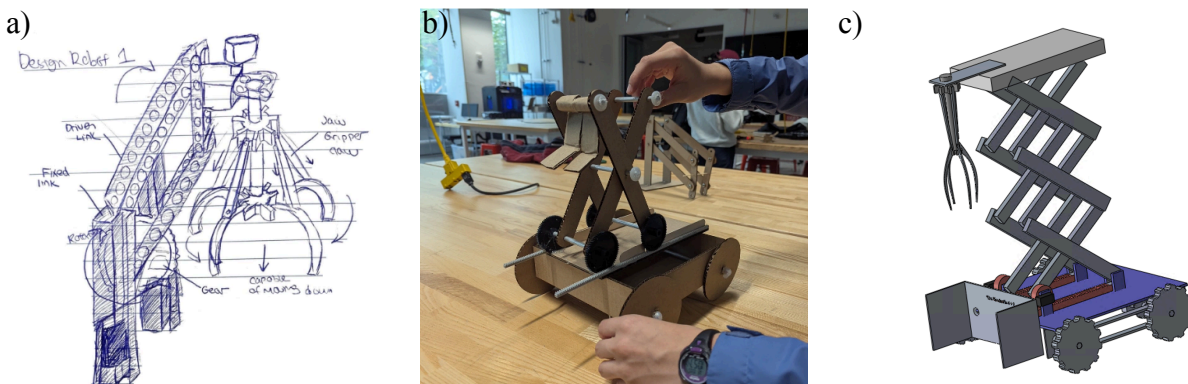


Figure 1.6: Scissor Lift Design, a) vertical claw sketch, b) cardboard prototype, c) CAD model

1.4b Four-Bar Linkage Arm Design

This concept consisted of a front shovel and a horizontal claw mounted on a four-bar linkage arm. Four-bar linkages are commonly used for their mechanical reliability and simplicity. However, analysis with the Pugh matrix revealed several limitations. The linkage's long horizontal sweep increased the risk of losing objects when lifting and the momentum of the arm produced unfavorable torque when lifting heavier objects. Additionally, even with a front shovel as a counterweight, calculations suggested stability concerns of the linkage on the 15° and 30° ramps. Although mechanically simpler than the scissor lift the four-bar linkage did not meet our criteria for precision, reach, and stability.

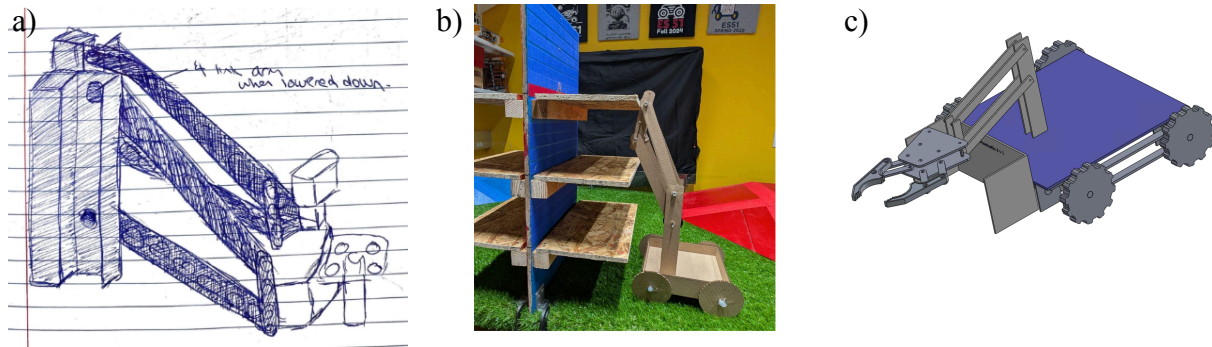


Figure 1.7: Four-Bar Linkage Design, **a)** four-bar linkage sketch, **b)** cardboard prototype, **c)** CAD model

1.5 Pugh Matrix

To select the final design, we created a Pugh decision matrix to evaluate each concept against important competition criteria. The telescoping linear slide design outperformed the alternative designs and was therefore selected as the final design direction.

Criteria	Weight	Design 1	Design 2	Design 3
Mobility	3	1	1	1
Collection speed	1	1	0	1
Scoring Speed	5	0	-1	1
Stability (CoG)	3	-1	1	-1
Turning radius	1	0	0	0
Turning speed	1	0	-1	0
Weight	2	1	-1	0
Durability	2	0	0	1
Fabrication Ease	2	1	-1	-1
Simplicity	3	1	0	1
<u>Total</u>		8	-4	9
Table 1: Pugh Chart Comparing Initial Three Designs, Design 1: Four Bar Linkage, Design 2: Scissor Lift, Design 3: Linear Slide				

2. Analysis

Motor Characteristics							
V (V)	I _{stall} (A)	R (Ω)	T _{stall} (Nm)	No-load RPM	No-load I (A)	K _t (Nm/A)	K _v (V/(rad/s))
1.2	4.4	0.54545	0.183	7.539	2.2	0.04159	0.15915
1.4	4.9	0.7	0.191	8.378	2	0.04898	0.16711
1.6	5	0.72727	0.214	9.425	2.2	0.0428	0.169765
Table 2: Motor Characteristics Chart containing values found in lab							

2.1 Drivetrain Calculations

The drivetrain calculations assessed whether the robot could climb the 15° and 30° ramps without exceeding traction or motor current limits. Using the robot's overall mass of 2.104 kg and wheel radius, 0.05m, along with motor constants, and estimated acceleration, we computed the maximum torque that could be applied before the wheels slipped. For the 15° ramp the required wheel torque was 0.160 Nm, with a motor current of 1.63 A. For the 30° incline the required wheel torque was 0.284 Nm and the motor current was 2.9 A. In both cases the required torque was less than the friction limit so the wheels did not slip and the motor current remained below the 9 A limit.

2.2 Linear Slide Calculations

The pulley plates of the linear slide with an area of $6.73 \times 10^{-4} \text{m}^2$ were evaluated under the combined slide, claw, and payload mass. The calculated shear stress of the bottom plate was 0.0075MPa. The calculated shear stress of the middle plate was 0.0063 MPa, and the calculated shear stress of the top plate was 0.0050 MPa. All stresses are far below the 30-60 MPa yield strength of PLA so the linear slide was structurally safe from any possible shearing.

2.3 Claw Grip and Torque Calculations

The heaviest payload, 0.125 kg for the dog toy, requires a minimum friction force of 0.65 N per claw. With a friction coefficient of 0.6, each claw must apply at least 1.08 N of normal force. This corresponds to a torque requirement of 0.163 N, compared to the servo motor's stall torque of 0.657 Nm at 6V. Since $0.163 \text{ Nm} < 0.657 \text{ Nm}$, the servo provides more than enough torque to grip and lift all objects.

2.4 Key Findings

Both inclines were climbable and within the traction and current limits for our drivetrain. The stresses acting on our linear slide were much lower than our material yield strength, and the claw had sufficient gripping torque to safely lift up the scoring objects.

3. Final Solution

The final robot integrated three major components: a four-wheel drivetrain, a three-stage telescoping linear slide, and a horizontal claw for object manipulation. Together, these components enabled our robot to navigate the field, collect objects, and score with precision on the shelves.

3.1 Functionality

The drivetrain uses a laser-cut baseplate, to which all components are mounted, and two sideplates held together with aluminum mounts to maintain wheel alignment. The robot has four wheels and uses rear-wheel drive powered with a 3:1 gearbox ratio. Although we initially planned to use a belt and pulley transmission, drivetrain testing revealed that the robot could climb the 15° and 30° ramps without it, so we removed the system to simplify the design. The 3.9" Ecoflex 30 molded wheels provided high traction while keeping the robot within the 11" height limit. The linear slide converts rotary motion into vertical motion using a belt-and-pulley cascade and is driven by a motor with a gearbox ratio of 1:6. Each of the three stages contains a top and bottom mount separated by two aluminum shafts, four rollers to decrease sliding friction, and top and bottom pulleys that are connected by belts. The belts are secured to the base plate of the slide via interference plates that ensure that the next stage is pulled upwards when the pulleys spin. The horizontal claw mounts to the final stage and uses a servo driven pair of 3D printed rippers with silicone inserts to increase friction.

3.2 Competition Performance

The robot passed all size and design constraints and performed well in the competition, placing third overall, and performing the best out of the non four-bar linkage designs. It did not malfunction on the field and reliably collected and placed the wooden blocks. The final design satisfied all key criteria by remaining compact, reliable, and mechanically unique.

3.3 Advantages and Disadvantages

Advantages of the robot included stable ramp climbing, a compact drivetrain, precise linear motion, effective gripping of objects, and overall reliability. Some disadvantages included the robot's inability to score tennis balls and dog toys, the high complexity of the CAD and design, the tight tolerances needed for the linear slide, and the bulky pulley mounts that required filing on competition day. Despite these limitations, the final design was solid and compact, satisfying our team's criteria by prioritizing reliability and creativity while meeting the Turf Wars competition constraints.

3.4 Future Improvements

With more time, we would reduce the size of the pulley mounts to prevent stage interference, increase the sides extension to reach the tennis ball shelf, and include a double row of rollers to increase stability of the individual stages and prevent tipping.

4. Final Design Specifications

DESCRIPTION	SPECIFICATION
Robot Mass	2.104 kg
Robot Base	10.5x10.5 in
Robot Height	11 in
Undercarriage Clearance	0.9 in
Turning Radius	0 in
Drivetrain Gear Ratio	3
Linear Slide Gear Ratio	1.6
Linear Slide Length Retracted	9.5 in
Linear Slide Length Extended	15 in
Wheel Base	6.5 in
Wheel Diameter	3.9 in
Wheel Coefficient of Friction	1.9
Claw Grip Coefficient of Friction	1.9
Number of Actuators	4
Maximum Claw Width	7 in

5. Appendix

5.1 Full Calculations

Torque calculations

$$TE = F_{RR} + F_{gx} + F_A$$

Variables			
K_T	Motor torque constant	τ_M	Motor torque needed for car to drive
$F_{RR} = 0$	Force necessary to overcome rolling resistance	TE	Tractive effort
F_{gx}	Gravitational force in the adjusted x-direction	F_f	Friction force by ground on each wheel
F_A	Force needed to accelerate the robot	τ_M	Torque exerted by motor
$m = 2.104kg$	Robot mass	I	Current supplied by motor
$m_w = \frac{2.104}{4} = 0.526kg$	Mass over the drive wheel	F_N	Normal force exerted by ground on each wheel
$\mu_{paint} = 1.9$	Coefficient of static friction between ecoflex 00-30 silicone and paint	$FS = 1.5$	Factor of Safety, approximated to 1.5
τ_w	Torque exerted per wheel	$GR = 3$	Gear Ratio from motor to wheel
τ_{max}	Maximum tractive torque that can be applied before slipping	$R = 0.05m$	Wheel radius
n	Number of independently powered wheels		
$\frac{v}{t_A} = 0.5m/s^2$	Desired robot acceleration		

Torque calculations for going up the 15° incline

$$F_{gx} = mgsin(\theta) = (2.104kg)(9.81m/s^2)sin(15^\circ) = 5.342N$$

$$F_A = \frac{mgv}{gt_A} = \frac{(2.104kg)(9.81m/s^2)(0.5m/s)}{(9.81m/s^2)(1s)} = 1.052N$$

$$\tau_w = \frac{TE \cdot R}{n} = \frac{mgR}{n} (sin\theta + \frac{v}{gt_a}) = \frac{(2.104kg)(9.81m/s^2)(0.05m)}{2} (sin15^\circ + \frac{0.5m/s}{9.81m/s^2(1s)}) = 0.15985Nm$$

$$\tau_{max} = F_f R$$

$$F_f = \mu F_N = \mu m_w gcos(\theta)$$

$$\tau_{max} = \mu m_w gcos(\theta)R = (1.9)(\frac{2.104}{4})(9.81m/s^2)cos15(0.05m) = 0.4735Nm$$

$$\tau_M = \frac{(FS)(\tau_w)}{GR} = \frac{(2)(0.15985 \text{ Nm})}{(3)} = 0.079925 \text{ Nm}$$

$$\tau_M = K_T I$$

K_T is taken as the K_T from voltage = 1.4V in the motor characterization. $K_T = 0.04898 \text{ Nm/A}$

$$I = \frac{0.079925 \text{ Nm}}{0.04898 \text{ Nm/A}} \approx 1.63 \text{ A} \rightarrow \text{Current draw does not exceed the maximum current draw}$$

$\tau_w < \tau_{max} \rightarrow$ Wheel will not slip as the robot goes up the 15° incline,

Torque calculations for going up the 30° incline:

$$F_{gx} = mgsin(\theta) = (2.104 \text{ kg})(9.81 \text{ m/s}^2)sin(30^\circ) = 10.32 \text{ N}$$

$$F_A = \frac{mgv}{gt_A} = \frac{(2.104 \text{ kg})(9.81 \text{ m/s}^2)(0.5 \text{ m/s})}{(9.81 \text{ m/s}^2)(1 \text{ s})} = 1.052 \text{ N}$$

$$\tau_w = \frac{TE \cdot R}{n} = \frac{mgR}{n} \left(sin\theta + \frac{v}{gt_a} \right) = \frac{(2.104 \text{ kg})(9.81 \text{ m/s}^2)(0.05 \text{ m})}{2} \left(sin30^\circ + \frac{0.5 \text{ m/s}}{9.81 \text{ m/s}^2(1 \text{ s})} \right) = 0.2843 \text{ Nm}$$

$$\tau_{max} = F_f R$$

$$F_f = \mu F_N = \mu m_w g cos(\theta)$$

$$\tau_{max} = \mu m_w g cos(\theta) R = (1.9) \left(\frac{2.104}{4} \right) (9.81 \text{ m/s}^2) cos30 (0.05 \text{ m}) = 0.4245 \text{ Nm}$$

$$\tau_M = \frac{(FS)(\tau_w)}{GR} = \frac{(1.5)(0.2843 \text{ Nm})}{(3)} = 0.14215 \text{ Nm}$$

$$\tau_M = K_T I$$

K_T is taken as the K_T from voltage = 1.4V in the motor characterization. $K_T = 0.04898 \text{ Nm/A}$

$$I = \frac{0.14215 \text{ Nm}}{0.04898 \text{ Nm/A}} \approx 2.9 \text{ A} \rightarrow \text{Current draw does not exceed the maximum current draw}$$

$\tau_w < \tau_{max} \rightarrow$ Wheels will not slip as the robot goes up the 30° incline

Linear Slide Stress Calculations

Linear Slide Stress Calculations	
Parameter	Kg/m
Arm Mass	0.261
Payload Mass	0.125
Gripper Mass	0.131
Pulley Plate Width	0.075692
Pulley Plate Thickness	0.00889

Variables Used	
σ	Shear Stress
F	Force parallel to cross-section area
A	Cross-section area of plate
a (g)	Gravitational acceleration

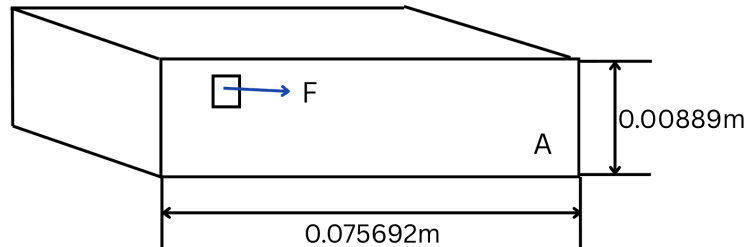


Figure 5.1: Representation of the volume of the linear slide plates

Pulley Bottom Plate:

$$\sigma = \frac{F}{A}$$

$$F = ma = mg = (0.261 + 0.125 + 0.131) \cdot 9.8 = 5.0666N$$

$$A = 0.00889 \cdot 0.075692 = 0.0006729m^2$$

$$\sigma = \frac{5.0666N}{0.0006729m^2} = 7529.5 Pa = 0.007529 MPa$$

Pulley Middle Plate:

$$\sigma = \frac{F}{A}$$

$$F = ma = mg = \left(\frac{2 \cdot 0.261}{3} + 0.125 + 0.131\right) \cdot 9.8 = 4.214N$$

$$A = 0.00889 \cdot 0.075692 = 0.0006729m^2$$

$$\sigma = \frac{4.214N}{0.0006729m^2} = 6262.4461 Pa = 0.0062624 MPa$$

Pulley Top Plate:

$$\sigma = \frac{F}{A}$$

$$F = ma = mg = \left(\frac{0.261}{3} + 0.125 + 0.131\right) \cdot 9.8 = 3.3614N$$

$$A = 0.00889 \cdot 0.075692 = 0.0006729m^2$$

$$\sigma = \frac{3.3614N}{0.0006729m^2} = 4995.393 Pa = 0.004995 MPa$$

Claw Torque Calculations

Variables used	
ΣF_y	Net force in the vertical direction
F_f	Friction force per claw on payload
W_1	Payload weight
μ	Static friction coefficient
F_N	Normal force per claw on payload
F_{App}	Force applied per claw on payload
τ	Torque

Minimum Force Necessary for Claw

$$\Sigma F_y = 0$$

$$F_f + F_f - W_1 = 0$$

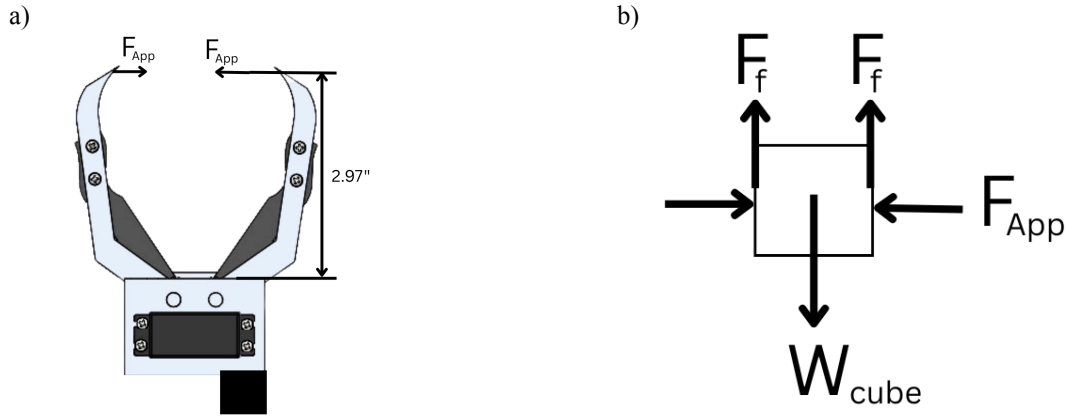


Figure 5.2: a) Claw Top View, b) Claw and Payload Free Body Diagram Front View

Torque Calculations

$$F_f = \frac{W_1}{2} = \frac{1.3N}{2} = 0.65N$$

$$F_f = \mu F_N = 0.6 \cdot F_{App} = 0.65N$$

$$F_{App} \geq 1.083N$$

$$\tau = F \cdot r = F_{App} \cdot 2.97'' \cdot \frac{0.0254m}{1''}$$

$$\tau_{load} = \tau_{N2} \cdot 2 = 0.16344 Nm$$

$$\omega_{no\ load, claw} = \frac{15^\circ}{1\ s} \cdot \frac{\pi\ rad}{180^\circ} = 0.262 \frac{rad}{s}$$

$$\tau_{stall, motor} = 6.7\ kg \cdot cm \cdot \frac{9.81N}{kg} \cdot \frac{1m}{100cm} = 0.6573Nm$$

$$\omega_{no\ load, motor} = 0.14 \frac{sec}{60^\circ} \cdot \frac{180^\circ}{\pi\ rad} = 0.13369 \frac{sec}{rad} = 7.48 \frac{rad}{s}$$

$$\tau_{load} < \tau_{stall, motor}$$

$$\omega_{no\ load, claw} < \omega_{no\ load, motor}$$

5.2 Figures

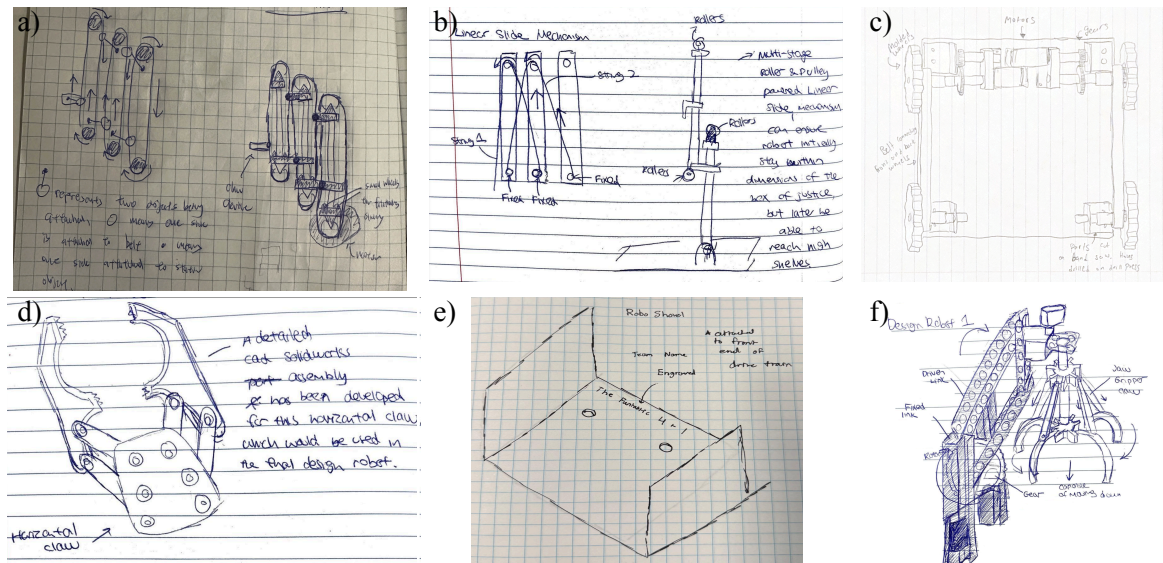


Figure 5.3: Initial Sketches, **a)** linear slide mechanism, **b)** extended linear slide mechanism, **c)** drivetrain, **d)** horizontal claw, **e)** shovel, **f)** vertical claw

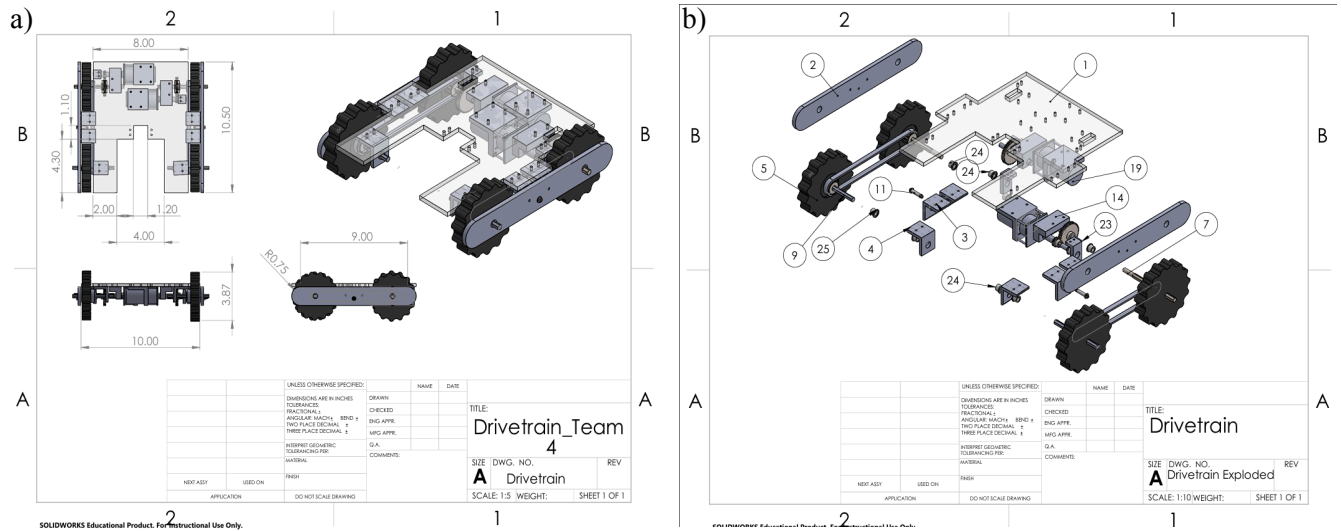


Figure 5.4: Drivetrain CAD, **a)** orthographic + isometric view, **b)** exploded view

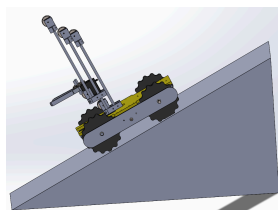


Figure 5.5: CAD Representation of Robot on Incline

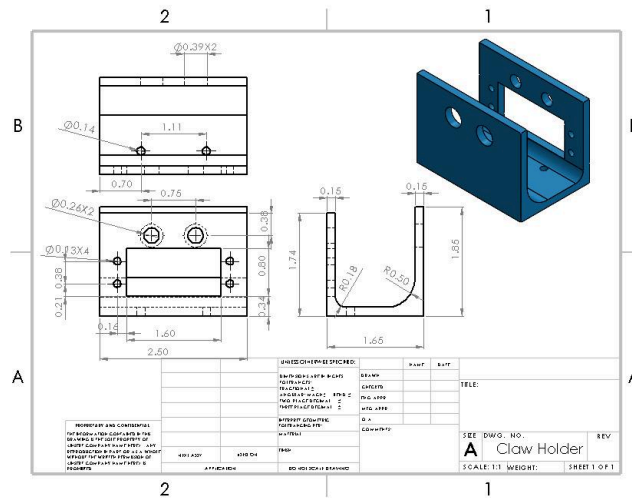


Figure 5.6: Drawing files of Claw Mount Piece

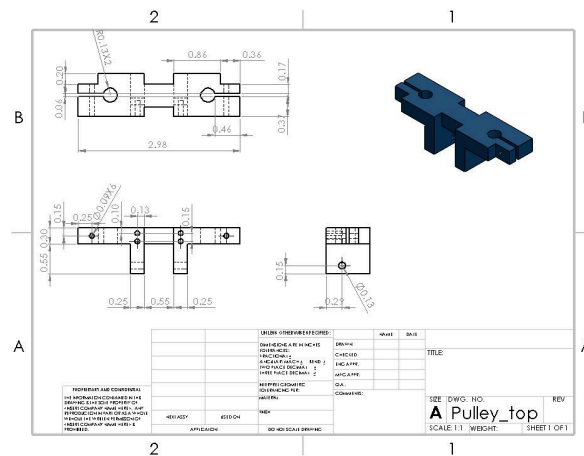


Figure 5.7: Drawing files of Linear Slide Top Piece

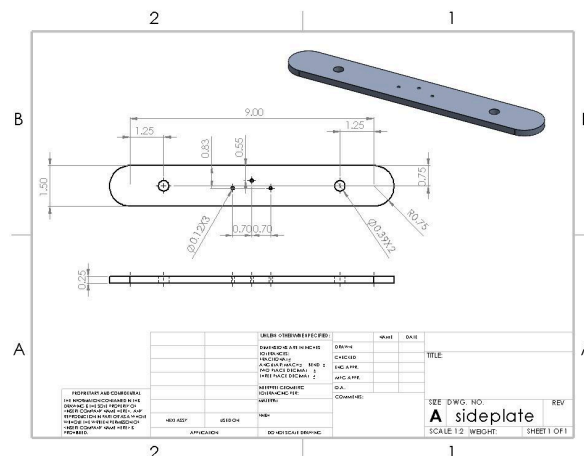


Figure 5.8: Drawing files of Drivetrain Sideplate

Bill of Materials

ITEM NO.	PART NAME	DESCRIPTION	QTY.
1	Attachment Plate	PLA 0.035 in ³	8
2	Claw Grip	PLA 1.196 in ³	4
3	Delrin Shaft Support	PLA 1.222 in ³	3
4	Initial Pulley Bottom Stage	PLA 2.109 in ³	1
5	Molded Claw Grip	Silicone EcoFlex 30	2
6	Motor Mount	PLA 1.3092 in ³	3
7	Pulley Bar Wheels	PLA 0.721 in ³	6
8	Pulley Bottom Stage	PLA 2.188 in ³	2
9	Smaller Gear Mount (Gearbox)	PLA 0.257 in ³	2
10	Spur Gear 32DP 16T 20PA 0.2FW	PLA 0.043 in ³	3
11	Spur Gear 32DP 24T 20PA 0.25FW	PLA 0.127 in ³	2
12	Spur Gear 32DP 44T 20PA 0.2FW	PLA 0.31 in ³	1
13	Spur Gear 32DP 48T 20PA 0.2FW	PLA 0.366 in ³	3
14	Top Pulley Stage	PLA 1.469 in ³	3
15	Upper Shaft Mount for Pulley	PLA 0.707 in ³	1
16	Wheel	Molded with Silicone EcoFlex 30	4

ITEM NO.	PART NAME	DESCRIPTION	QTY.
1	Baseplate	Acrylic ¼ inch	1
2	Timing Belt, .200" Pitch, 18" Outer Circle, 169K117	—	3
4	External Retaining Ring	—	10
5	Flat Head Phillips 4-40, ½ 9023A110	—	13
6	Front Wheel Mount	6061 Aluminum	2
7	Hex Shaft	6061 Aluminum	10
8	Locknut 4-40 90631A005	—	52
9	Motor	*Electric Screwdriver Workshop	3
10	Nylon Bearing ¼"	—	14
11	Nylon Unthreaded Spacer #4 Screw 94639A662	—	12
12	Pan Head Phillips 4-40, 1-½ 912A119	—	63
13	PowerHD 6001 HB Servo Motor	—	1
14	Pulley Bar	6061 Aluminum	6
15	Side Plate Mount	6061 Aluminum	4
16	Sideplate	Acrylic ¼ inch	2
17	Timing Belt Pulley	—	9

6. Contributions

Susanna: Scripted and Edited Team Video, Abstract, Put together the final report, edited all writing to fit technical guidelines, formatted figures. Organized all writings into distinct sections

Winfred: Led Final CAD Assembly, Section 5: Part drawings, Assembly Drawing, Section 1.3A Linear slide , Final Solution sections 3.1 & 3.2

Eva: Led Assembly of robot, Section 3: Final solutions, Section 1.5 Pugh Matrix, Section 4: Final Design Specifications

Serena: Led Claw CAD design and robot calculations, Appendix 5.1 and Part 2 Analysis (motor characteristics, torque going up 15° and 30° incline, linear slide arm stress calculations, claw torque, free body diagrams), 1.4b four-bar linkage design sketch, 1.4a scissor lift design sketch, Appendix 5.2 robot exploded view CAD.

Diego: Led Robot Testing, Section 1: Concept Development Section, Section 5: Bill of Materials,